

Bayesian Food Recommendation System

Abstract

Method: This project proposes a **Hybrid Bayesian Personalized Meal Recommendation System** that combines collaborative filtering with Bayesian inference to generate adaptive and personalized recipe recommendations. The system predicts the probability that a user will enjoy a recipe based on available ingredients, dietary restrictions, cuisine preferences, cooking time, nutritional information, and previous recipe ratings. The proposed methodology uses a hybrid recommendation framework in which the **Singular Value Decomposition (SVD)** algorithm first generates recommendation scores by learning patterns from historical user–recipe interactions. Bayesian inference then updates these predictions by refining prior beliefs into posterior probabilities as users provide new ratings and feedback. This approach is appropriate because collaborative filtering captures similarities between users, while Bayesian inference enables continuous personalization and provides interpretable confidence estimates for each recommendation.

Model: The project will be implemented in **Python** using the **Surprise** library for collaborative filtering and **PyMC** for Bayesian modeling and inference. The project will **not** leverage a pre-trained language model. Instead, the recommendation system will be developed using the SVD algorithm from the Surprise library together with Bayesian updating implemented in PyMC.

Data: The project will use the **Food.com Recipes and User Interactions** dataset from Kaggle, a real-world dataset containing over **180,000 recipes** and approximately **700,000 user ratings and reviews**. The dataset includes recipe metadata, ingredients, nutritional information, cooking instructions, cuisine and dietary tags, and historical user–recipe interactions, providing sufficient information for both collaborative filtering and Bayesian updating.

Dataset:

https://www.kaggle.com/datasets/shuyangli94/food-com-recipes-and-user-interactions?select=RAW_recipes.csv